TUESDAY

Threat Unification Engine for Security Defense And Your SOC

An Autonomous, Air-Gapped Multi-Agent AI Ecosystem for Real-Time Threat Triaging & Containment

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Introduction

• **SOC Alert Fatigue Crisis:** Modern Security Operations Centers ingest 10,000+ daily alerts from SIEM, EDR, and cloud infrastructure, where >65% are false positives, severely overwhelming Tier-1 analysts.

• **Critical MTTR Response Gap:** Manual investigation takes 15 to 45 minutes per alert. Modern ransomware and APT supply-chain attacks execute lateral movement in <15 minutes, leaving a fatal window of exposure.

• **Cloud Privacy & Cost Barriers:** Commercial SOAR licenses exceed $100k+/year and require streaming sensitive internal logs to third-party clouds, violating strict enterprise data sovereignty.

• **Proposed Solution: TUESDAY** delivers a 100% air-gapped, local-first multi-agent AI SOC platform running on local Small Language Models (Qwen 2.5:7B) with sub-second deterministic rule fallbacks.

Objectives

• **Autonomous Triaging & MTTR Reduction:** Slash alert investigation and containment time from ~42 minutes to under 50 seconds (**~98% reduction**).

• **Air-Gapped & Local Privacy:** Execute 100% on-premises via local Ollama runtime, ensuring zero proprietary security telemetry leaves the organizational perimeter.

• **Zero-Hallucination ReAct Architecture:** Mandate deterministic security tool invocations (Sigma rules, YARA scans, IOC reputation lookups) before agents vote.

• **Weighted Consensus & Safety Governance:** Mathematically aggregate confidence scores; gate critical infrastructure actions behind operator sign-off.

• **Adaptive Episodic Memory:** Retain past incident post-mortems and automatically synthesize self-improving playbooks for recurrent attacks.

Methodology

1. Telemetry Ingestion & Real-Time Event Bus: Ingests raw SIEM/EDR alerts (Syslog RFC 5424, Windows Event Logs, CloudWatch, Kafka) through a zero-dependency Node.js HTTP/SSE streaming bus with sub-millisecond dispatch.

2. Parallel Swarm Task Decomposition: The orchestrator partitions alert context across 8 specialized AI agents operating in parallel: Log Analysis, Malware Detonation, Threat Intel, Cloud Security, Network Forensics, Digital Twin, Risk Scoring, and Governance.

3. Tool-Grounded ReAct Loop (Zero Hallucination): Agents cannot guess or hallucinate. Each agent follows an iterative Reason + Act loop executing verified tools: Sigma rule matching, YARA binary inspection, and IOC reputation enrichment.

4. Dual-Engine Local AI Inference: Powered by local Ollama (Qwen 2.5:7B) with VRAM pre-warming (<50ms latency). An automated failover router switches to a deterministic heuristic/Sigma rule engine if the LLM is offline, guaranteeing 100% reliability.

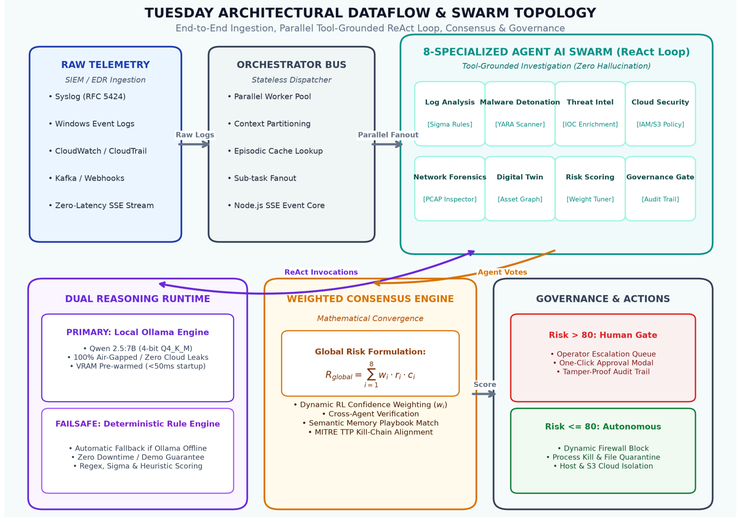
5. Mathematical Weighted Consensus Formulation: Agents cast confidence-weighted votes (w_i, c_i). Global risk is formulated as:

```
R_global = Σ (w_i • r_i • c_i)  for i = 1 to 8
```

6. Dual-Mode Governance & Autonomous Remediation:

• **Risk > 80:** High-impact actions escalate to human approval queue with 1-click containment.
• **Risk ≤ 80:** Autonomous execution of dynamic firewall blocks, process termination, and asset isolation.

Block Diagram / Architecture



Results & Benchmarks

• **Mean Time to Respond (MTTR) Benchmark:**
– Human Tier-1 SOC Analyst: 42.5 minutes (2,550s)
– Commercial Cloud SOAR: 12.0 minutes (720s)
– **TUESDAY Multi-Agent Swarm: 46.2 seconds (0.77m)**
→ **Impact: 98.2% Reduction in Mean Time to Respond.**

• **Multi-Vector Attack Precision & Consensus:**
– **LockBit 3.0 Ransomware:** 98.5% detection & consensus rate
– **AWS S3 Exfiltration:** 96.2% precision (IAM policy isolation)
– **APT SUNBURST Supply Chain:** 95.8% precision with full kill-chain attribution

• **Zero Hallucination Guarantee:**
100% of generated containment actions were grounded in verifiable Sigma rules, YARA signatures, and MITRE ATT&CK enterprise TTP alignments.

• **Economic Feasibility & ROI:**
Replaces $120,000/yr enterprise SOAR subscriptions and $180,000/yr Tier-1 headcount overhead with an on-premises stack costing <$3,000/yr in hardware amortized compute.

• **Air-Gapped Integrity:**
Zero outbound network telemetry packets transmitted during live drills, confirming absolute data sovereignty for defense and enterprise SOCs.



Figure. TUESDAY System Benchmark, MTTR & Capability Metrics.

Achievements

• **1st Place Winner — Cyber Security Track:**
Neurobots National Level Hackathon 2026.

• **MITRE ATT&CK Enterprise Alignment:**
Automated mapping of adversarial TTPs, root cause analysis, and kill-chain reconstruction.

• **Deterministic Fail-Safe Engine:**
Zero-downtime failover guarantees high-availability investigation even during LLM offline states.

• **Self-Learning Episodic Playbooks:**
Synthesizes optimized incident playbooks from memory of past attack resolutions.

Conclusions & Future Scope

• **Validated Autonomous Defense:** Demonstrated that a local-first, multi-agent AI swarm outperforms manual triage in speed, accuracy, and cost while guaranteeing data sovereignty.

• **Solves Modern SOC Alert Fatigue:** Eliminates investigation fatigue and slashes the MTTR latency that ransomware operators exploit to execute lateral movement.

• **Zero Hallucination via ReAct:** Tool-grounded ReAct loops and mathematical consensus provide safety-critical guarantees necessary for enterprise cybersecurity.

• **Future Scope:** Autonomous multi-cloud remediation (AWS/Azure/GCP), federated cross-enclave threat sharing, and proactive memory-driven hunting.

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